

COSC-1437 Programming Fundamentals II

TARRANT COUNTY COLLEGE DISTRICT MASTER SYLLABUS

COURSE DESCRIPTION

This course focuses on the object-oriented programming paradigm, emphasizing the definition and use of classes along with fundamentals of object-oriented design. The course includes basic analysis of algorithms, searching and sorting techniques, and an introduction to software engineering processes. Students will apply techniques for testing and debugging software. This course is included in the Field of Study Curriculum for Computer Science.

Prerequisite: COSC-1436

COURSE TYPE

Academic Non-Core

COURSE GOALS AND LEARNING OUTCOMES

Upon successful completion of this course, students will:

1. Identify and explain a programming development lifecycle, including planning, analysis, design, development, and maintenance.
2. Demonstrate a basic understanding of object-oriented programming by using structs and classes in software projects.
3. Use object-oriented programming techniques to develop executable programs that include elements such as inheritance and polymorphism.
4. Document and format code in a consistent manner.
5. Apply basic searching and sorting algorithms in software design.
6. Apply single- and multi-dimensional arrays in software.
7. Use a symbolic debugger to find and fix runtime and logical errors in software.
8. Demonstrate a basic understanding of programming methodologies, including object-oriented, structured, and procedural programming.
9. Describe the phases of program translation from source code to executable code.

COURSE ASSESSMENT

Student success is measured by a variety of assessment techniques aligned with course goals and learning outcomes. Individual faculty members are responsible for designing evaluation instruments to measure student mastery of course goals and learning outcomes and indicating the nature of such evaluation instruments in the instructor's class requirements.

STUDENT ACCESSIBILITY RESOURCES

Any student with a documented disability needing academic accommodations is required to contact the Student Accessibility Resources (SAR) Office located on each campus to schedule an appointment with the Coordinator of SAR. All discussions are confidential. Because SAR accommodations may require early planning and are not provided retroactively, students are encouraged to contact SAR as early in the semester as possible. SAR is responsible for approving and coordinating all disability-related services. TCC professors will honor requests for accommodation when they are issued by SAR.

SCHOLASTIC DISHONESTY

Students are responsible for adhering to the TCCD policy on scholastic dishonesty as stated in the online student handbook at www.tccd.edu.

RESOURCES

For additional information regarding the student handbook, academic calendar, course evaluations, attendance policy, SCANS skills, Core Competencies, etc., see www.tccd.edu.